Gunath Kalpitha

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Profile

Computer Engineering undergraduate at the Faculty of Engineering, University of Ruhuna, with a strong focus on **Artificial Intelligence** and **Machine Learning**, complemented by a multidisciplinary background in **IoT** and **Full Stack Development**. Passionate about building intelligent, data-driven systems using **PyTorch**, **Scikit-learn**, and **Python**, with hands-on experience in computer vision, predictive modeling, neural networks, and natural language processing. Experienced in developing secure mobile applications using Flutter and scalable backend systems using Java and Node.js. Driven to engineer end-to-end AI-powered solutions that solve real-world problems.

Education

BSc (Hons) in Computer Engineering (Undergraduate)
Faculty of Engineering, University of Ruhuna 2023 – Present
Current GPA: 3.1

Rajapaksha Central College, Weeraketiya 2018 – 2021

Projects

Digital Inspection System for National Building Research Organization (Ongoing)
Flutter, Supabase, PostgreSQL, Google Maps API
Project Link: Source Code

- Built a cross-platform digital inspection and pre-crack survey mobile application for NBRO infrastructure projects.
- Developed an offline-first Flutter application for GPS-tagged field data and photo capture in low-connectivity environments.
- Integrated Supabase Authentication, PostgreSQL, and Edge Functions for secure real-time and delayed synchronization.

Sri Lanka Law Chatbot – RAG-Powered Legal Assistant
Python, LangChain, ChromaDB, Gemini API, FastAPI, React.js
Project Link: Source Code

- Building a Retrieval-Augmented Generation (RAG) legal chatbot using LangChain, ChromaDB, and Gemini API.
- Enables users to query official Sri Lankan legal documents and acts with accurate cited responses.
- Implemented document ingestion using PyMuPDF and vector search with ChromaDB.
- Developed backend services with FastAPI and frontend interfaces using React.js and Tailwind CSS.

Project Sentry (2025) – IoT Smart Home Energy Management System
ESP32, React.js, MongoDB, Node.js, Express.js
Project Link: Source Code

- Participated in the design and development of an IoT-based smart home system for energy management and safety.
- Developed a web dashboard using React.js and Tailwind CSS for real-time monitoring.
- Integrated MongoDB Cloud for energy consumption tracking and data storage.
- Selected as one of the Top 8 finalist teams in FINNC25 organized by the IEEE Student Branch, University of Moratuwa.

Sleep-Aware Smart Alarm System (2026) – Ongoing
Python, Scikit-learn, Random Forest, Pandas, NumPy
Project Link: Source Code

- Developing a sleep-stage classifier using Random Forest algorithms on wearable physiological signal datasets.
- Engineered 12 features from 30-second epochs across 2,800+ labeled samples.
- Utilized PPG, HRV, and Accelerometer signals to classify Wake, Light Sleep, and Deep Sleep stages.

Technical Skills

Programming Languages	Python, Java, C#, JavaScript, Dart
AI / ML	PyTorch, Scikit-learn, LangChain, ChromaDB, NLP, Computer Vision
Frontend	React.js, Tailwind CSS, HTML, CSS, Flutter
Backend	Node.js, Express.js, RESTful APIs
Databases	MongoDB, MySQL, PostgreSQL
Cloud Platforms	Firebase, Supabase
DevOps & Tools	AWS, Docker, Jenkins, Terraform, Git
Professional Skills	Teamwork, Time Management, Problem Solving, Fast Learner

Certificates

- Ballerina Competition
- AI/ML Engineer Stage 1
- Web Development Certification

Extra-Curricular Activities

- Organizing Committee Member, Rextro 2025 Silver Jubilee Exhibition, Faculty of Engineering, University of Ruhuna.
- Member, Gaveshakayo Hiking Club – Participated in faculty-organized hiking events demonstrating teamwork, planning, and endurance.

References

Dr. Kushan Sudheera
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